Machine Learning — Short Notes

1. What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence where computers learn patterns from data and use them to make predictions or decisions without explicit programming for every case.
Example: Train on spam/not-spam emails and predict a new email's class.

2. Types of Machine Learning

Supervised: learns from labelled data — Regression, Classification.
Unsupervised: finds patterns in unlabelled data — Clustering, PCA.
Semi-supervised: combines a small labelled dataset with a large unlabelled dataset.
Reinforcement: learns using rewards and penalties.

3. Common ML Problems

Regression: predicts a continuous number, such as house price.
Classification: predicts a class, such as spam/not spam.
Clustering: groups similar observations without predefined labels.

4. Typical ML Workflow

Define problem → collect data → clean data → EDA → feature engineering → split data → train model → evaluate → tune → deploy → monitor.

5. Data Preparation

Handle missing values; encode categorical variables; scale features when appropriate; inspect outliers; remove irrelevant or duplicate data; avoid data leakage.

6. Train / Validation / Test

Training: model learns from data.
Validation: model/hyperparameters are selected and tuned.
Test: final unbiased evaluation on unseen data.

7. Popular Algorithms

Linear Regression: continuous prediction.
Logistic Regression: classification.
Decision Tree: classification/regression.
Random Forest: ensemble of trees.
Gradient Boosting: strong tabular-data performance.
KNN/SVM: classification and regression.
K-Means: clustering.
PCA: dimensionality reduction.
Neural Networks: complex patterns, images and text.

8. Evaluation Metrics

Regression: MAE, MSE, RMSE, R^2.
Classification: Accuracy, Precision, Recall, F1-score, ROC-AUC.
Clustering: Silhouette Score and related measures.

9. Overfitting vs Underfitting

Overfitting: performs very well on training data but poorly on unseen data. Reduce using simpler models, regularization, more data, feature selection, cross-validation, etc.
Underfitting: model is too simple and performs poorly on both training and unseen data.

10. Parameters vs Hyperparameters

Parameters: learned from training data, such as model weights.
Hyperparameters: chosen before/during training, such as learning rate, tree depth or number of neighbors.

11. Bias and Variance

High bias: model too simple → underfitting.
High variance: model too sensitive to training data → overfitting. Good models seek a suitable balance.

12. Cross-Validation

K-Fold Cross-Validation divides training data into K parts. The model trains on K–1 parts and validates on the remaining part, repeating until every part has been used for validation. Scores are then averaged.

13. Feature Engineering

Creating, transforming or selecting useful input variables. Examples: extracting month/day from dates, creating age from date of birth, log-transforming skewed values, or selecting important features.

14. ML vs Deep Learning vs Generative AI

ML: broad family of algorithms that learn from data.
Deep Learning: ML based on multi-layer neural networks, especially useful for complex unstructured data.
Generative AI: models that generate new text, images, audio, code and other content.

15. Key Terms

Dataset = collection of data | Feature = input variable | Target/Label = value to predict | Model = learned representation | Training = learning from data | Inference = prediction on new data | Loss = prediction error | Optimizer = method for updating model parameters.

Quick Learning Order

Python → NumPy/Pandas → Statistics & Probability → EDA → Supervised Learning → Unsupervised Learning → Evaluation → Feature Engineering → Hyperparameter Tuning → ML Projects → Deep Learning.